Build with AI

Dubai, UAE

Building And Deploying An Agentic AI App With Google Cloud And ADK

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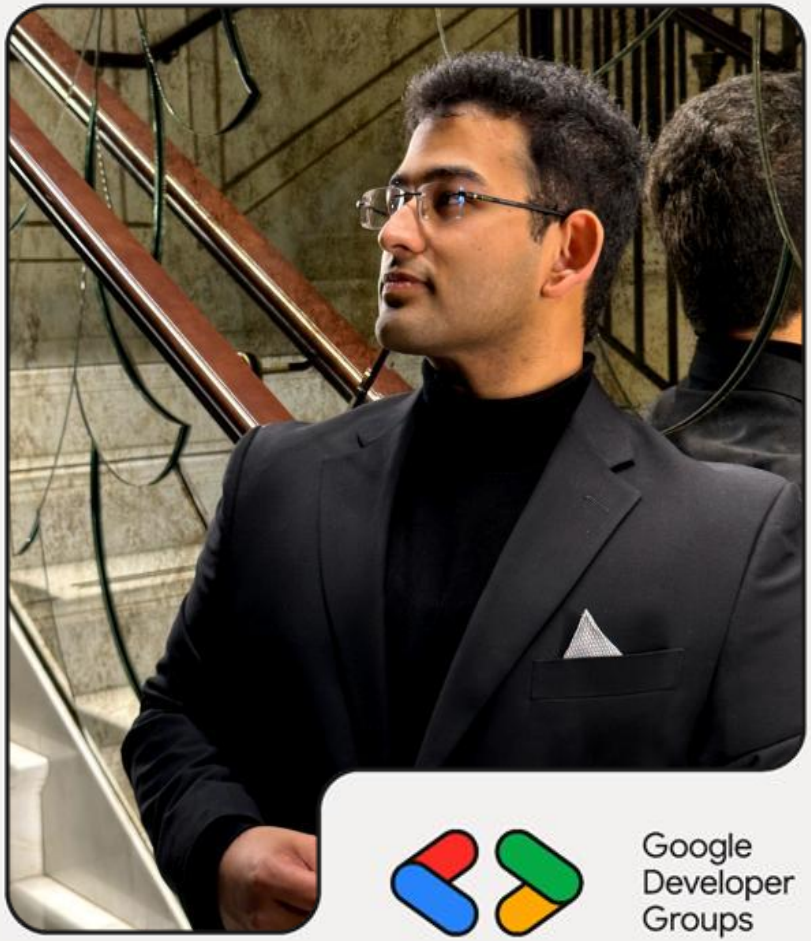

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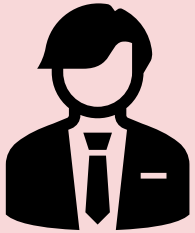
LLMs vs Agentic AI

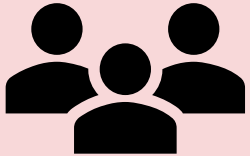
LLM

- Reasoning and knowledge, but cant take actions
- Stateless
- Passive



Agent

- Body = tools
- Job Desc = instructions & goals
- Autonomous
- Stateful
- Take actions till job is done



Multi-Agent

- Talented and focused individuals
- Get tasks to help achieve goal
- Focused
- Modular, reliable, debug-able

Why ADK?

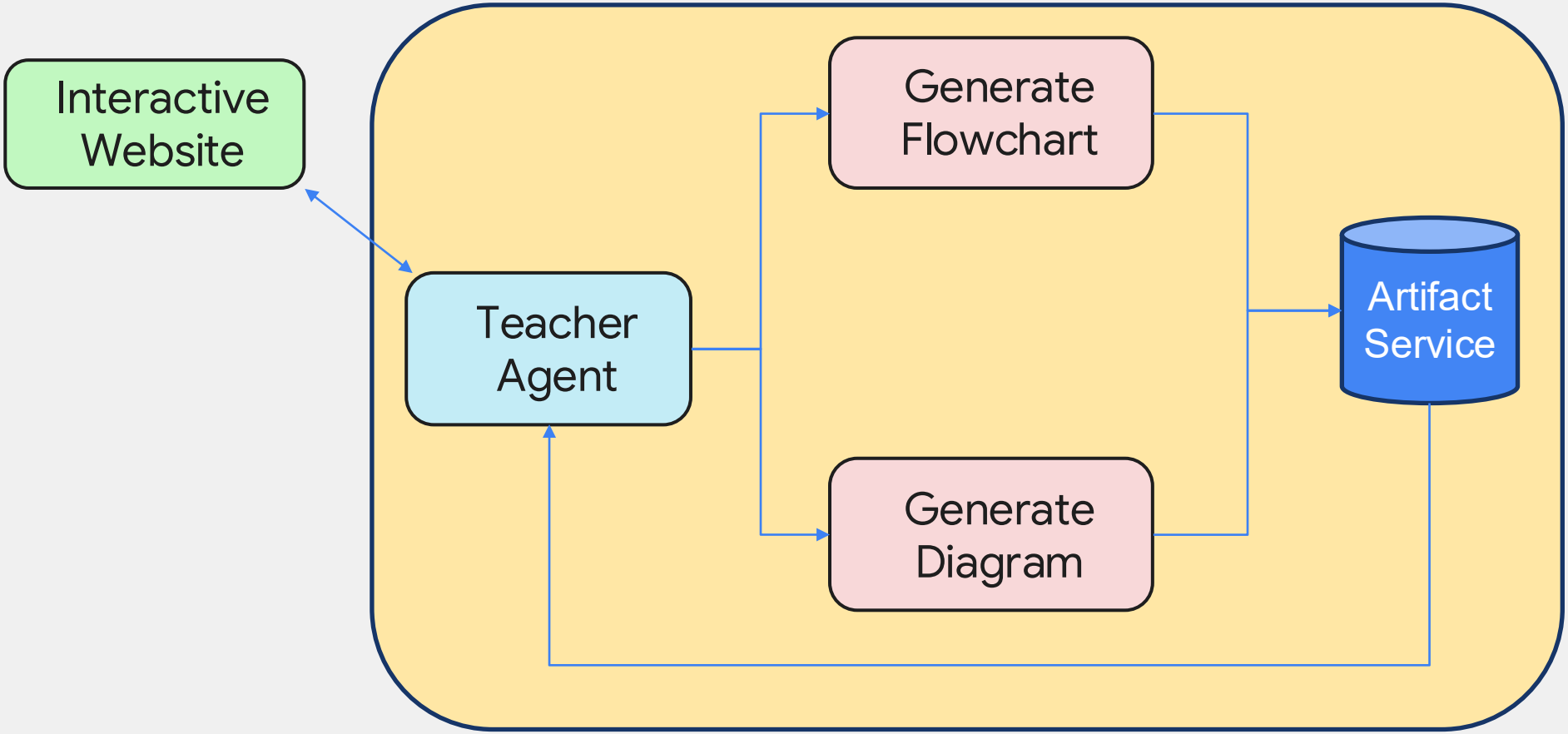
- Need control to build complex agents and workflows
  - Modularity
  - Reusability
  - State management
- Easy deployment and integration
- Quick set up while being scalable

Case Study – Research Explainer

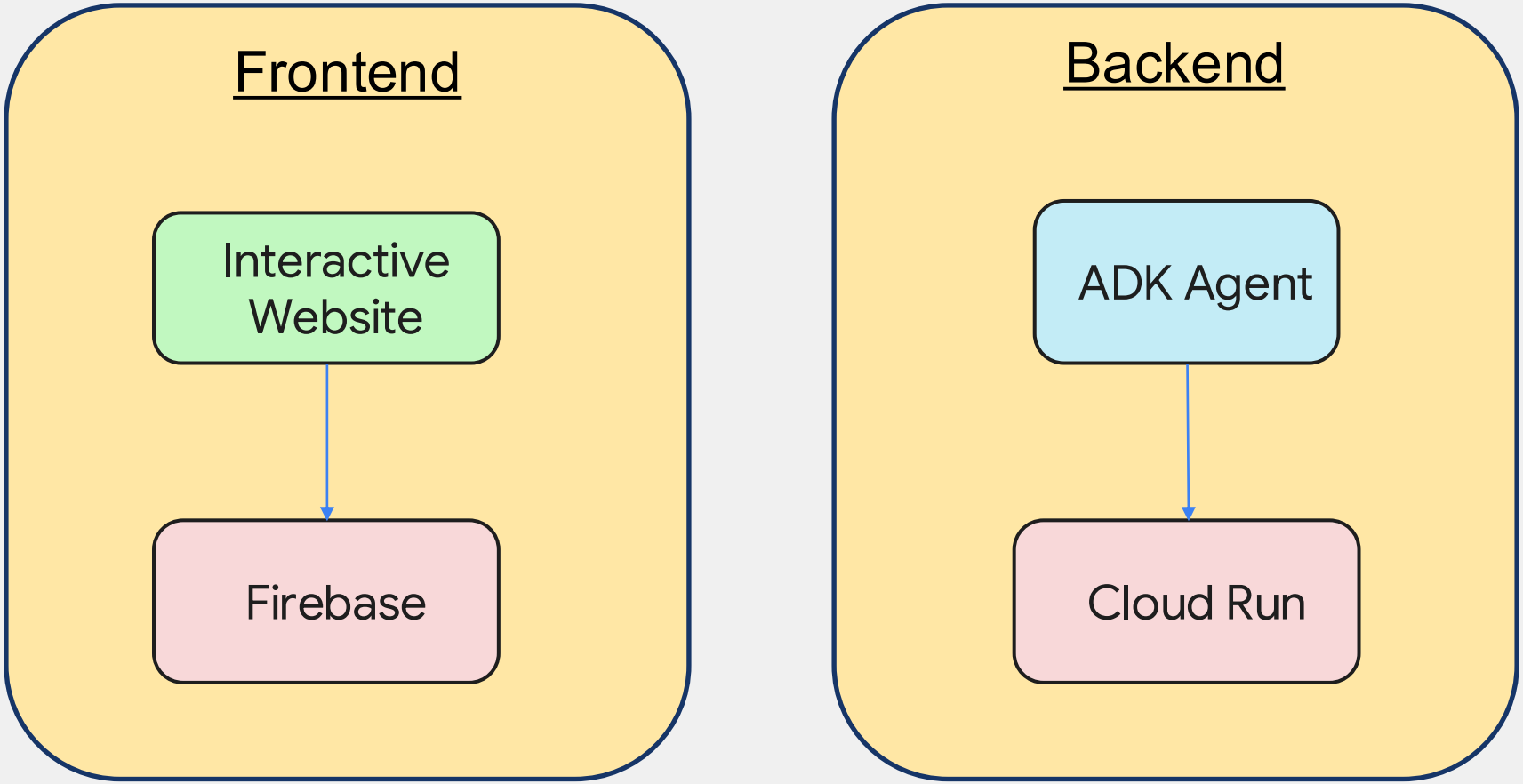
- Take a research paper / technical document and explain it while generating required diagrams and charts

- One Agent to read and understand the paper
- Tool call to generate flow charts – show processes
- Tool call to generate diagrams and images – show concepts

Our Architecture



Our Deployment Strategy



Artifact & Sessions

Artifacts

- Any large object that is loaded/created
- Includes files, images, data, etc
- **Artifact service:** A temporary storage space for the session
- Easy way to transfer large objects between agents
- Saves to RAM / GCS / Redis

Sessions

- Agent's memory of this conversation
- Handled by ADK
- **Session service:** Where the session details will be saved (temp)
- Important when multiple agent pods exist – share the same memory
- Saves to RAM / GCS / Redis

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Demo
Code on Github

github.com/ro1406/research-paper-explainer

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Deployment

- Deploy directly to GCP – Cloud Run
- Non-GCP deployments – Docker container
  - Single service
  - Host as `adk api_server –port 8000`
- Identify good Artifact services
  - RAM – local / low traffic / single deployment
  - GCS good for GCP deployments
  - Redis sidecar for kubernetes deployments
- Serverless deployments
  - Event driven
  - Scalable (to 0 if needed)

Medium.com Article

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Thank You!

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